AP BIOLOGY TOPIC 6.3

Transcription, RNA Polymerase & Eukaryotic Pre-mRNA Processing

UNIT 6: GENE EXPRESSION & REGULATION (12–16% WEIGHT)

IST-1.N | TEMPLATE 3′→5′ | 5′ CAP & POLY-A TAIL | ALTERNATIVE SPLICING

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 6.3 (TRANSCRIPTION, RNA POLYMERASE & EUKARYOTIC PRE-MRNA PROCESSING)

CED Core Vocabulary

Central Dogma of Molecular Biology — Genetic information flows directionally from DNA (permanent blueprint) → RNA (transient intermediate via Transcription) → Polypeptide / Protein (functional...

Coding Strand — (also called Non-template, Plus (+), or Sense Strand): Runs 5′ → 3′ ; has the exact same sequence and polarity as the newly synthesized mRNA except every T is...

Three Stages of Transcription — Initiation: RNA Polymerase binds to the promoter DNA sequence upstream of the gene (in eukaryotes, general transcription factors must first bind the promoter's...

Termination — In prokaryotes, stops at a terminator hairpin sequence; in eukaryotes, RNA Pol II transcribes a polyadenylation signal (AAUAAA) and releases the pre-mRNA...

Template Strand — (also called Noncoding, Minus (–), or Antisense Strand): The ONLY DNA strand physically read by RNA Polymerase in the 3′ → 5′ direction to build a complementary...

RNA Polymerase Properties — Unwinds the DNA double helix locally by itself (no separate helicase needed) and does NOT require an RNA primer to initiate synthesis.

Elongation — RNA Polymerase moves 3′ → 5′ along the template strand , elongating the RNA transcript 5′ → 3′ (with U pairing with A); DNA rewinds behind it.

3′ Poly-A Tail (50–250 Adenines) — Added to the 3′ end by poly-A polymerase ; increases mRNA stability/lifespan in the cytosol and facilitates nuclear export.

§2. CENTRAL DOGMA & FUNCTIONAL CLASSES OF RNA

DNA → RNA → Protein

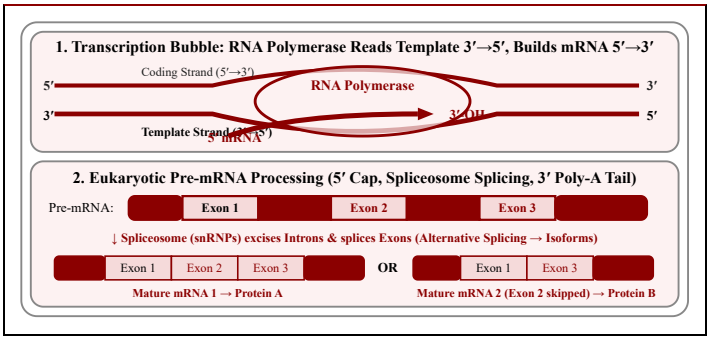
- **Central Dogma of Molecular Biology:** Genetic information flows directionally from **DNA** (permanent blueprint) → **RNA** (transient intermediate via *Transcription*) → **Polypeptide / Protein** (functional machinery via *Translation*).

RNA TYPE	SPECIFIC MOLECULAR ROLE IN GENE EXPRESSION
mRNA (Messenger RNA)	Single-stranded transcript carrying triplet **codons** from DNA to the ribosome to specify the linear amino acid sequence of a polypeptide.
tRNA (Transfer RNA)	Cloverleaf/L-shaped adaptor molecule carrying a specific **amino acid at its 3′ end** and a complementary **3-base anticodon loop** that base-pairs with mRNA codons.
rRNA (Ribosomal RNA)	Complexes with ribosomal proteins to form the **large and small ribosomal subunits**; acts as a catalytic **ribozyme** (peptidyl transferase) to form peptide bonds!

[COMMON AP EXAM FRQ TRAP] **Don't Swap mRNA and tRNA Roles:** **mRNA** carries the genetic message (codons); **tRNA** physically delivers amino acids (via anticodons) to the ribosome!

§4. TRANSCRIPTION & EUKARYOTIC PRE-MRNA PROCESSING VISUALIZED

Introns Out, Exons In



§5. EUKARYOTIC PRE-MRNA PROCESSING & PROKARYOTE COMPARISON

Eukaryotes Only!

HIERARCHICAL CONCEPT & PATHWAY MAP | [CAUSE ➔ MECHANISM ➔ OUTCOME]

Eukaryotic Pre-mRNA Processing & Prokaryote Comparison ➔ **Eukaryotes** ➔ **Alternative RNA Splicing**

- In **Eukaryotes**, primary transcripts (**pre-mRNA**) undergo **3 mandatory covalent modifications** inside the nucleus before export to the cytoplasm:
  1. **5′ Methyl-GTP Cap (5′ G-P-P-P):** Added to the 5′ end; protects mRNA from hydrolytic 5′ exonucleases, aids export through nuclear pores, and helps the **small ribosomal subunit recognize & bind** the 5′ end.
  2. **3′ Poly-A Tail (50–250 Adenines):** Added to the 3′ end by *poly-A polymerase*; increases mRNA stability/lifespan in the cytosol and facilitates nuclear export.
  3. **RNA Splicing (via the Spliceosome / snRNPs):** Excises non-coding intervening sequences (**Introns**) and splices together coding expressed sequences (**Exons**) to form continuous mature mRNA.
- **Alternative RNA Splicing:** Different combinations of exons can be included or skipped from the same primary transcript → allows **a single eukaryotic gene to code for multiple distinct polypeptide isoforms** in different tissues!
- **Prokaryotes vs. Eukaryotes (High-Yield Distinction):**
  - **Prokaryotes:** Lack a nuclear envelope and lack introns → **NO RNA processing**; ribosomes attach to mRNA while it is still being transcribed (**coupled / simultaneous transcription & translation** in the cytoplasm).
  - **Eukaryotes:** Nuclear envelope spatially & temporally separates **transcription + RNA processing (in nucleus)** from **translation (in cytoplasm / rough ER)**.

[CORE AXIOM / REMEMBER] **Memory Hook: EXons** are **EXpressed** (and **EXit** the nucleus in mature mRNA); **INtrons** are **INtervening** non-coding sequences that stay **IN** the nucleus (cut out and degraded)!

§3. TEMPLATE VS. CODING STRAND & 3 STEPS OF TRANSCRIPTION

Reads 3′→5′, Builds 5′→3′

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

Template vs. Coding Strand & 3 Steps of Transcription ➔ **Template Strand** ➔ **Coding Strand** ➔ **RNA Polymerase Properties** ➔ **Three Stages of Transcription**

- **Template Strand** (also called *Noncoding, Minus (–), or Antisense Strand*):
  - The **ONLY** DNA strand physically read by **RNA Polymerase** in the **3′ → 5′ direction** to build a complementary **5′ → 3′ RNA transcript** (**A→U**, **T→A**, **G↔C**).
- **Coding Strand** (also called *Non-template, Plus (+), or Sense Strand*):
  - Runs **5′ → 3′**; has the **exact same sequence and polarity as the newly synthesized mRNA** except every **T is replaced by U**!
- **RNA Polymerase Properties:**
  - Unwinds the DNA double helix locally by itself (no separate helicase needed) and **does NOT require an RNA primer** to initiate synthesis.
  - Synthesizes RNA strictly **5′ → 3′** by adding ribonucleoside triphosphates (NTPs) to the 3′-OH end.
- **Three Stages of Transcription:**
  1. **Initiation:** **RNA Polymerase** binds to the **promoter** DNA sequence upstream of the gene (in eukaryotes, general **transcription factors** must first bind the promoter's **TATA box** to recruit RNA Pol II); DNA unwinds into a transcription bubble.
  2. **Elongation:** RNA Polymerase moves **3′ → 5′ along the template strand**, elongating the RNA transcript **5′ → 3′** (with **U pairing with A**); DNA rewinds behind it.
  3. **Termination:** In prokaryotes, stops at a terminator hairpin sequence; in eukaryotes, RNA Pol II transcribes a polyadenylation signal (**AAUAAA**) and releases the **pre-mRNA** transcript.

[EXAMPLE & BIOLOGICAL APPLICATION] **Instant Strand Conversion**

Practice:

- **DNA Coding (5′→3′):** **5′ – A T G C C T A G T – 3′**
- **DNA Template (3′→5′):** **3′ – T A C G G A T C A – 5′**
- **mRNA Transcript (5′→3′):** **5′ – A U G C C U A G U – 3′**

[COMMON AP EXAM FRQ TRAP] **Always Check Which Strand Is Given!** If an exam problem gives you the **Coding (5′→3′) strand**, simply swap **T → U**. Only take the complement if given the **Template (3′→5′) strand**!



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§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 6.3 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 6.3: Transcription, RNA Polymerase & Eukaryotic Pre-mRNA Processing ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 8 & 9 (pGLO Bacterial Transformation & Restriction Enzyme Analysis of DNA): Heat-shock transformation of E. coli with pGLO plasmid (bla ampicillin-resistance gene + GFP under arabinose-inducible araC promoter). Plate comparisons: −pGLO/LB (lawn, viability control), −pGLO/LB+Amp (no growth, antibiotic lethality control), +pGLO/LB+Amp (white colonies, transformation), +pGLO/LB+Amp+Ara (glowing green colonies under UV, inducible gene expression). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative & Graph Analysis: Transformation Efficiency = (Total colonies counted on selective plate) / (µg of plasmid DNA spread). Gel Electrophoresis Semi-Log Standard Curve: DNA fragment size (bp, $\log_{10}$ scale on y-axis) vs. migration distance (mm, linear scale on x-axis) yields a straight line for interpolating unknown fragment lengths. PCR amplification after n thermal cycles yields 2^n target DNA copies. **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Molecular Disruption Case Study: Splice-site mutations in the β-globin gene (β-Thalassemia) prevent the spliceosome (snRNPs) from excising an intron or cause exon skipping, shifting the mRNA reading frame and introducing a premature nonsense stop codon (UAA/UAG/UGA). In oncology, epigenetic hypermethylation of CpG islands in tumor-suppressor promoters (e.g., BRCA1 or MLH1) or histone deacetylation condenses euchromatin into transcriptionally silent heterochromatin without altering the primary DNA sequence.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 6.3 to prove why the claim holds.